

DS 3010 Study Guide

Study Resources

1. Homeworks + homework solutions
2. Midterm 1 and Midterm 2
3. Your lecture notes and R scripts
4. In-class activities
5. Textbook

Topics (in chronological order)

Be thoughtful and strategic in your approach to studying. Many of the methods are constructed from the same building blocks. Focus on making sure your understanding of those building blocks are solid.

1. Multiple Linear Regression

- Setup the model. What are the true population parameters? What do they represent? What is the irreducible error? What does it represent?
- How do we estimate parameters in this model conceptually? What do we call these estimates?
- What are some properties of these estimates ($\hat{\beta}$)?
- When does the method do particularly well?

2. Inference for Multiple Linear Regression and Multiple Testing Problems

- What assumption(s) are needed just to fit a multiple linear regression model?
- Confidence intervals vs. prediction intervals - what's the difference?
- What is the multiple testing problem? Can you simulate data setting to demonstrate this problem?
- What is the F-test? Explain conceptually what components are needed for an F-test.

3. Bias-variance tradeoff

- What is the definition of bias and variance in conceptual terms plain language?
- For all supervised learning methods, how does bias and variance behave as you increase model flexibility?
- For simulated data, how would you write code to calculate bias square and variance for simulated data?
- In regularized regression, when λ increases, how do bias/variance behave?
- In K-NN classification, when K increases, how do bias/variance behave?

- In decision trees, when α increases, how do bias/variance behave?
- How does training error behave as you increase model flexibility?

4. Model Selection

- Describe model selection techniques available to us mathematically and conceptually - subset selection, greedy algorithms, subset + cv, and other variations.
- What are the strengths and limitations of each approach?
- Can you use RSS for model selection? Explain why or why not.
- Be sure to understand coding/implementation details. For example, is AIC computed on the full dataset or training set only?
- How do we protect against double-dipping?

5. MLR other considerations

- Categorical predictors - how do we incorporate them into the model?
- Interaction terms - when do we need them? How do we decide if we want to include them in the model?
- Polynomial regression - how do we decide which degree to use?
- Model diagnostics
- Multicollinearity - when is it a problem?

6. Regularization: Ridge and lasso

- What is the motivation for shrinkage methods?
- Setup the models. How do we estimate parameters in regularized models conceptually?
- When do the methods do particularly well?
- Distinction between ridge and lasso?
- Coding - what is your workflow? How do you protect against double-dipping?

7. Resampling methods: cross-validation and bootstrap

- Be sure to understand implementation + coding details very well.
- We have seen cross-validation in almost every module. What is the purpose of cross-validation?
- What are the limitations/advantages of different flavors of cross-validation validation set approach, LOOCV, k-fold CV?
- What is the purpose of bootstrapping? When would you use bootstrap instead of other methods?
- How do you code bootstrapping for standard errors and confidence intervals?

8. Classification

- What models are available to carry out classification?
- Setup the models. If needed, how do we estimate parameters in these models conceptually?
- When do the methods break down?
- When do the methods do particularly well, i.e. under what scenario would we choose to use LDA over logistic regression? Under what scenario would we choose to use KNN over Naive Bayes?
- ROC curves, false positive/false negatives, threshold tuning

9. Tree-based Methods

- Basic decision trees - how are they fit? How does recursive binary splitting work?
- Tree-cost complexity pruning: what is it? How is it implemented? How does it tie into bias-variance tradeoff?
- Classification trees
- Bagging + random forest: setup the model. Understand the algorithm to ensemble the trees. What are the tuning parameter(s) and how do we tune them? Statistical justification for bagging and random forest?
- Boosting: setup the model. Understand the algorithm to ensemble the trees. What are the tuning parameter(s) and how do we tune them?